

KRUSHNA SUTARIYA

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ABOUT ME

Software Engineer with hands-on experience in React, TypeScript, Python, Docker, CI/CD, Git, and cloud-based application development. At BMW Group, built real-time visualization tools for ADAS and localization data using React, TypeScript, Mapbox GL JS, Python, WebSockets, Chart.js, Plotly, and Foxglove Studio. Recently submitted an M.Sc. thesis on explainable LLM-based recommendation systems and completed academic cloud/DevOps projects using AWS, GCP, Terraform, and GitHub Actions. Strong problem solver, quick learner, and motivated to contribute to reliable full-stack and cloud-enabled applications.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java basics

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Mapbox GL JS, React Leaflet, Chart.js, Turf.js, UI/UX

Backend and APIs: Python, REST APIs, WebSockets, SQL Server, DynamoDB

Cloud and DevOps: AWS, GCP Cloud Run, Docker, Terraform, GitHub Actions, GitLab CI/CD, Git, Linux, pytest

AI and Machine Learning: LLM concepts, Explainable AI, recommender systems, Scikit-learn, Pandas, NumPy

Tools: Foxglove Studio, Figma, Jira, Postman, Google Colab

PROFESSIONAL EXPERIENCE

BMW Group

Software Development Intern

Feb 2025 – Jul 2025

Munich, Germany

- Redesigned the web-based Localization and Map Visualization platform for BMW's ADAS team using React.js, TypeScript, and Mapbox GL JS, including modular UI components, collapsible sidebars, layer controls, and dynamic map-state handling, resulting in a 40% increase in data analysis speed for test-drive validation teams.
- Built dual-layer geospatial visualizations for lane and road position candidates using Mapbox GL JS and Turf.js, including color-coded paths and timestamp hover popups.
- Developed a Foxglove Studio configuration panel with React.js and TypeScript for live layer visibility control of traffic signs, lane markings, centerlines, and map settings.
- Replaced static PDF optimization plots with a real-time browser dashboard using Python, WebSockets, Plotly, Matplotlib, and Chart.js frontend visualizations.
- Implemented 2D and 3D vehicle localization hypothesis visualizations, including WGS84-to-ENU coordinate conversion and probability-scaled cylinders in Foxglove Studio.
- Worked in an agile Scrum environment with sprint planning, daily stand-ups, retrospectives, Git version control, and GitHub pull request reviews.

Techsphere Softwares LLP

Front-End Developer

Jan 2021 – Apr 2023

India

- Developed responsive web pages and reusable UI components using HTML, CSS, and JavaScript for client-facing applications.
- Integrated frontend views with backend APIs and supported usability improvements, browser compatibility testing, and clean code maintenance.
- Collaborated with developers and stakeholders to translate requirements into functional web interfaces.

Savaj Infotech

Web Designer

Aug 2016 – Dec 2020

India

- Designed and implemented website layouts, landing pages, and UI components with focus on usability, responsiveness, and visual consistency.
- Created UI concepts and page structures for client projects and collaborated with developers to deliver functional web solutions.

PROJECTS AND RESEARCH

SkyMap Weather Application

React.js, Vite, Tailwind CSS, React Leaflet, OpenWeather API, GitHub Pages

Jun 2026

- Built and deployed a responsive weather map application with city search, interactive map selection, live weather data, and custom weather UI components.
- Integrated external APIs and map-based interactions to display weather, forecast details, location-based data, and user-friendly visual feedback. Live demo: krushnasutariya.github.io/sky-map-weather

Master's Thesis: Explainable LLM-Based Recommendation System

Nov 2025 – Apr 2026

Hochschule Heilbronn, Germany

- Designed explainable recommendation strategies for the IdeaLize academic collaboration platform using textual, keyword-based, and visual explanation formats.
- Compared textual, keyword-based, and visual explanation formats to improve transparency and user trust.
- Built a high-fidelity Figma prototype and conducted a user-study-based evaluation to analyze trust, clarity, usefulness, and user perception of explanations.
- Applied Explainable AI principles, recommender-system concepts, and human-centered interface design.

DevOps and SecOps CI/CD Automation Project

Academic Project

Python, Docker, GitHub Actions, pytest, Google Cloud Run

- Containerised a Python REST API and configured automated CI/CD workflows with GitHub Actions, pytest, and code coverage checks.
- Deployed the application to Google Cloud Run using an automated image build and cloud deployment workflow.

Serverless Online Grocery Application

Academic Project

AWS, Terraform, Lambda, S3, DynamoDB, API Gateway, GitLab CI/CD

- Built a serverless web application architecture on AWS using Terraform for infrastructure provisioning and automated deployment.
- Implemented IAM, S3, DynamoDB, Lambda, and API Gateway, supported by a GitLab CI/CD pipeline.

Movie Recommendation System

Academic Project

Python, Pandas, NumPy, Scikit-learn, Google Colab

- Built a recommendation system using collaborative and content-based filtering concepts to generate personalized movie suggestions.
- Processed datasets and evaluated recommendation logic using data preprocessing and similarity-based matching.

EDUCATION

Hochschule Heilbronn

Mar 2024 – Expected 2026

Master of Science in Software Engineering

Germany

Current average grade: 1.7; thesis submitted Apr 2026

Relevant coursework: Advanced Software Architecture, DevOps and SecOps, Cloud Computing, Designing User Interfaces, Cybersecurity

Shree Swami Atmanand Saraswati Institute of Technology (SSASIT)

Jun 2013 – May 2016

Bachelor of Engineering in Computer Engineering

India

CGPA: 8.36 / 10

Government Polytechnic for Girls

Jun 2010 – May 2013

Diploma in Computer Engineering

India

CGPA: 8.64 / 10

LANGUAGES

English - Fluent C1 | German - A2 / B1 intensive course in progress | Gujarati - Native | Hindi - Native

CERTIFICATIONS

AWS IoT | DevOps on AWS | Fundamentals of Machine Learning and AI | Intermediate Machine Learning